

ECCA Roadside Litter Study

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Overview

- Sampling Design (Mark)
- Categorization Design (Addison)
- Collection Protocol (Maddi)
- Categorization Protocol (Khloud)

ECCA Roadside Litter Study

Sampling Design

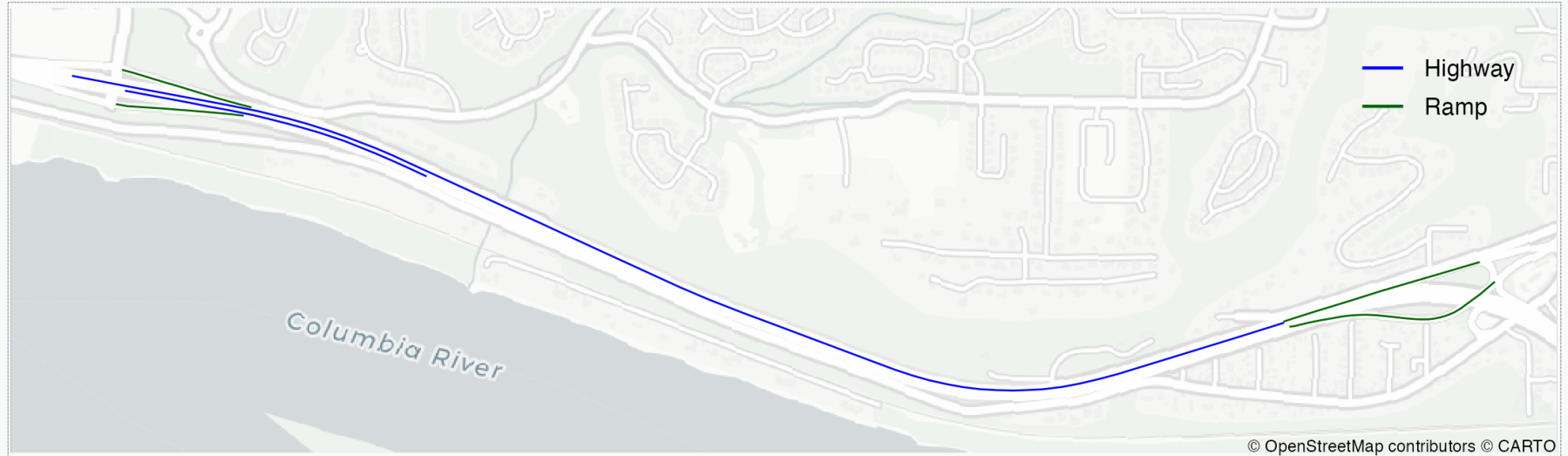
Approach

Determine areas (strata) to use to group all the highways and ramps

Assume trash in a given area is relatively similar anywhere in the area, but possibly different from other areas

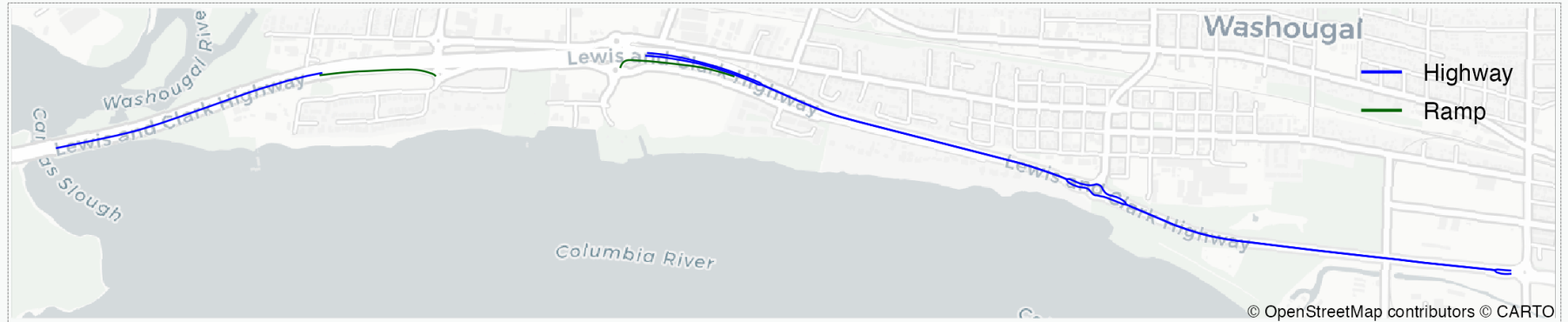
Pick some number of segments from each area to use in the sample

West Side Areas



- 2 highway stretches (north/westbound, south/eastbound)
- 2 pairs of ramps

East Side Areas



- 3 highway stretches
- 1 pair of ramps

Sampling Execution

Idea

Do sampling entirely in software using geospatial data from WSDOT

We use data from the **WSDOT Geospatial Open Data Portal**:

- Highway data: **WSDOT - State Route Linear Referencing System (LRS) Current**
- Ramp data: **WSDOT - State Route Ramp (1:24K) Current**

Highway Sampling

1. Load geospatial data for both sides of Highway 14

2. Filter these into the 5 areas using the milepost boundaries for what ECCA cleans

3. Divide each area into 600 foot segments

4. Randomly select one of these segments in each area

Ramp Sampling

Idea

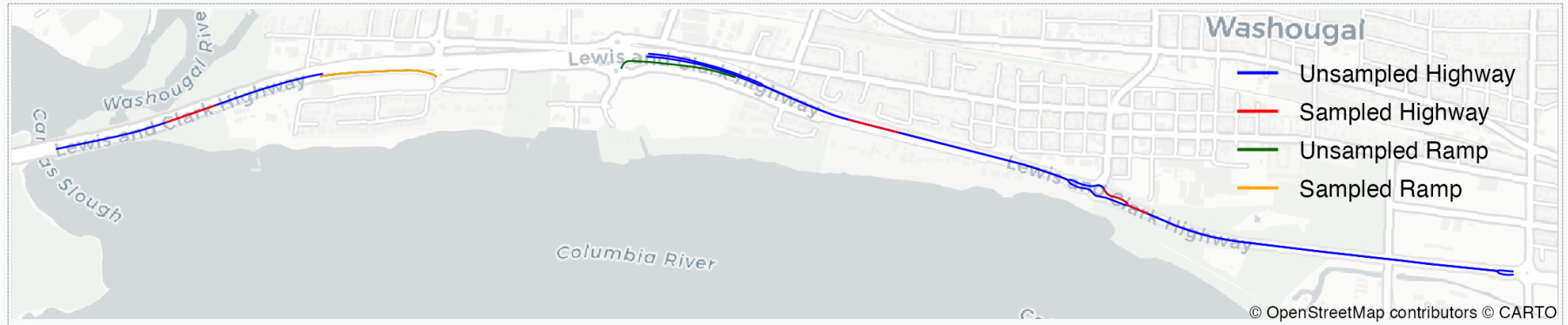
Pick one ramp from each pair, but make sure at least one on-ramp and one off-ramp are selected

Sampling Results: West Side



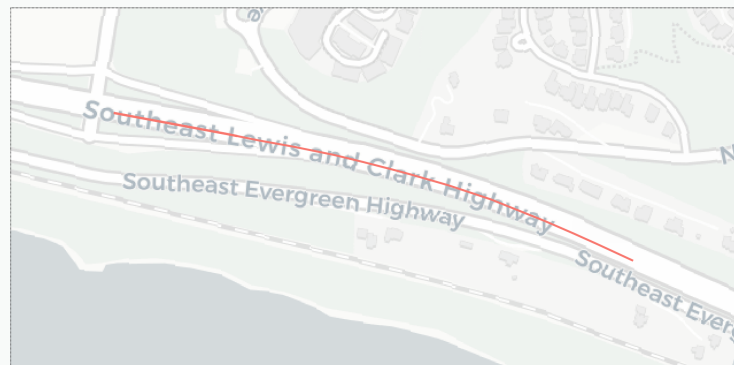
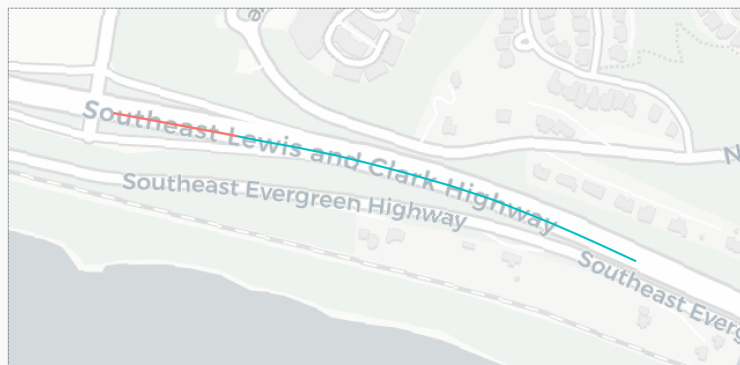
- 2 highway samples
- 2 ramp samples

Sampling Results: East Side



- 3 highway samples
- 1 ramp sample

Example Estimation: Plastic Bottles



100 plastic bottles found

x

4.3

=

430 plastic bottles estimated

(sample weight)

Categorization Design

MATERIAL (MUTUALLY EXCLUSIVE — PRIMARY MATERIAL BY VOLUME)

Paper / Cardboard

Plastic

Glass

Metal

Wood

Textile

Other

FUNCTIONAL CATEGORIES (SELECT ALL THAT APPLY)

Food Container

Toxic Fluid Container

Automotive Debris

Construction / Demolition

Lawn & Garden

Marina-Related

Attire

Furniture

Lumber Tag

Table 72. Detailed litter composition table for Interstate roadways.

Material	Est. %	+/-	Est. Pounds per Mile	Pounds per Mile +/-	Est. %	+/-	Est. Pounds per Mile	Pounds per Mile +/-	Est. Pounds per Mile	Pounds per Mile +/-
Paper	4.2%	0.6%	59.1	9.4	15.4%	1.2%	4,374	478	191.5	8,367
Cardboard	0.2%	0.2%	24.2	3.2	0.2%	0.2%	881	29	11.2	1,111
Fast-food Paper Bags	0.0%	0.0%	1.1	0.2	0.0%	0.0%	79	18	1.0	46
Fast Food Paper Bags	0.0%	0.0%	8.4	0.6	0.0%	0.0%	229	60	2.0	90
Fast-food Paper Cups	0.4%	0.1%	1.1	4.2	0.4%	0.1%	164	10	4.0	169
Cups	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	1.2	42
One-Trial To-Go/Fast Food Service Items	0.1%	0.0%	1.5	0.6	0.1%	0.1%	393	30	3.8	138
Paper Napkins and Paper Plates	0.1%	0.0%	1.6	0.6	0.1%	0.1%	218	20	8.5	1,444
Paper Menus and Food Cartons, Alcoholic	0.0%	0.0%	0.1	0.1	0.0%	0.0%	3	3	0.2	5
Paper Beverages and Food Cartons, Non-Alcoholic	0.0%	0.0%	0.2	0.1	0.0%	0.0%	23	0.5	0.5	297
Paper Non-Beverage Paper/Fast Food Packaging	0.1%	0.1%	3.1	0.8	0.0%	0.0%	259	76	2.9	149
Non-food Paper Packaging	0.1%	0.1%	3.1	1.4	0.1%	0.1%	261	76	7.4	446
Newspapers, Magazines, Leaf Mail, and Office Paper	0.2%	0.2%	7.7	2.7	0.2%	0.2%	1,498	267	164	1,110
House jbs	0.0%	0.0%	0.1	0.1	0.1%	0.1%	36	22	17.1	739
Other Paper	0.2%	0.2%	82.0	1.9	12.2%	0.8%	1,219	207	16.1	2,739
Plastics	13.1%	1.5%	151.0	18.2	38.6%	2.9%	14,844	1,519	118.1	18,770
Plastic Bottles and Lids, Beer	0.0%	0.0%	3.9	0.0	0.0%	0.0%	7	0	0.4	0
Plastic Bottles and Lids, Soda	0.0%	0.1%	2.4	0.7	0.0%	0.1%	195	40	2.9	92
Plastic Bottles and Lids, Single-serve Wine and Liquor	0.1%	0.0%	1.1	0.4	0.0%	0.1%	92	36	0.8	21
Plastic Bottles and Lids, Other Wine and Liquor	0.0%	0.0%	0.4	0.2	0.0%	0.0%	8	5	1.1	7
Plastic Bottles and Lids, Sports and Health Drinks	0.0%	0.0%	0.4	0.2	0.0%	0.0%	34	12	2.4	14
Plastic Bottles and Lids, Juice	0.2%	0.1%	2.4	1.0	0.2%	0.1%	66	28	0	0
Plastic Bottles and Lids, Tea and Coffee	0.0%	0.0%	0.3	0.2	0.0%	0.0%	14	7	1.2	12
Plastic Bottles and Lids, Still-water	0.4%	0.1%	4.7	0.9	1.2%	0.2%	479	69	8.8	414
Plastic Bottles and Lids, Other water	0.0%	0.0%	0.1	0.1	0.0%	0.0%	3	2	1.9	30
Plastic Bottles and Lids, Unknown	0.0%	0.0%	4.2	2.9	0.1%	0.0%	27	18	1.1	18
Plastic Cans and Lids	0.0%	0.0%	0.1	0.1	0.0%	0.1%	282	18	6.1	170
Plastic Bottle Caps and Tabs	0.0%	0.0%	0.2	2.5	1.0%	0.2%	346	40	2.3	440
Plastic Beverage Rings	0.0%	0.0%	0.0	0.0	0.0%	0.0%	3	3	0.1	7
Fast-food Plastic Caps	0.2%	0.1%	3.5	1.1	0.0%	0.1%	218	42	3.5	105
Plastic Caps	0.1%	0.0%	0.7	0.3	0.1%	0.1%	93	73	0.8	102
Plastic Lids	0.1%	0.1%	1.7	0.6	0.1%	0.2%	284	79	0.1	3
Plastic Unusable	0.0%	0.0%	0.4	0.4	0.1%	0.1%	98	47	0.2	26
Plastic, Resins and Single-use Containers	0.0%	0.0%	0.9	0.0	0.0%	0.0%	0	0	1.5	96
Clay/brick	0.0%	0.0%	0.1	0.1	0.0%	0.0%	0	0	0.0	0
Plastic Trash Bags	0.0%	0.1%	2.3	1.0	0.2%	0.0%	95	28	2.4	120
Grocery, Retail and Shopping Bags	0.1%	0.0%	0.1	0.4	0.1%	0.1%	10	12	1.2	134
Other Plastic Film	1.1%	0.3%	18.7	1.5	30.3%	1.2%	1,869	461	9.0	2,411
Food Wrappers and Trash Bags	0.1%	0.1%	8.9	1.2	1.6%	0.1%	1,191	199	7.9	1,311
Food and Drink Packaging	0.0%	0.0%	0.1	0.1	0.0%	0.0%	16	7	0.5	82
Food Cans, Resins, and Composites	0.0%	0.0%	9.8	0.3	0.0%	0.0%	167	0.5	0.5	82
Other Resins	0.0%	0.2%	9.8	2.9	4.2%	0.2%	1,611	268	8.7	1,618
Non-Beverage Plastic/Food Packaging	0.2%	0.2%	2.2	0.7	0.7%	0.2%	387	20	2.1	120
Non-food Plastic Packaging	0.2%	0.1%	3.7	1.2	0.0%	0.2%	125	48	12.4	2,345
Other Plastics	0.1%	1.1%	73.4	225	12.4%	1.4%	4,778	561	46.2	8,479
Glass	1.3%	0.4%	16.1	4.8	1.9%	0.8%	511	123	38.8	1,587
Glass Bottles and Lids, Beer	0.1%	0.0%	8.8	4.1	0.2%	0.1%	15	30	18.1	161
Glass Bottles and Lids, Soda	0.0%	0.0%	0.9	0.0	0.0%	0.0%	0	0	0.0	0
Glass Bottles and Lids, Single-serve Wine and Liquor	0.0%	0.0%	0.1	0.0	0.0%	0.0%	0	2	0.0	0
Glass Bottles and Lids, Other Wine and Liquor	0.0%	0.0%	0.1	0.0	0.0%	0.0%	0	0	0.0	0
Glass Bottles and Lids, Sports and Health Drinks	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	0.0	0
Glass Bottles and Lids, Juice	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	0.0	0
Glass Bottles and Lids, Tea and Coffee	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	0.0	0
Glass Bottles and Lids, Still-water	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	0.0	0
Glass Bottles and Lids, Unknown	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	0.0	0
Glass Bottles and Lids, Other Water	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	0.0	0
Other Glass/Food Packaging	0.1%	0.1%	1.1	0.8	0.1%	0.1%	30	19	2.1	86
Other Glass/Non-Food Packaging	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	0.0	0
Other Glass or Composites	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	0.0	0
Other Glass	0.1%	0.1%	3.5	1.7	0.1%	0.2%	253	19	4.1	1,110
Metal	6.1%	1.0%	83.5	22.8	5.1%	0.6%	1,977	238	18.1	3,438
Metal Bottles, Lids and Cans, Beer	0.4%	0.1%	4.7	1.1	1.0%	0.1%	491	118	8.7	144
Metal Bottles, Lids and Cans, Soda	0.2%	0.0%	1.9	0.1	0.0%	0.1%	189	10	2.0	96
Metal Bottles, Lids and Cans, Sports and Health Drinks	0.1%	0.1%	2.1	1.1	0.7%	0.1%	161	27	1.2	120
Metal Bottles, Lids and Cans, Juice	0.0%	0.0%	0.1	0.0	0.0%	0.0%	8	2	0.1	5
Metal Bottles, Lids and Cans, Tea and Coffee	0.0%	0.0%	0.1	0.2	0.1%	0.0%	46	1.0	1.0	25
Metal Bottles, Lids and Cans, Still-water	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	0.0	0
Metal Bottles, Lids and Cans, Other Water	0.1%	0.0%	1.2	0.9	0.0%	0.0%	12	0.3	0.3	12
Metal Bottles, Lids and Cans, Unknown	0.0%	0.0%	0.2	0.1	0.1%	0.0%	29	11	0.2	10
Non-Beverage Metal/Food Packaging	0.1%	0.1%	2.9	0.8	0.0%	0.0%	12	2.8	1.8	80
Metal Non-Food Packaging	0.1%	0.0%	0.4	0.4	0.1%	0.0%	30	18	1.4	443
Stainless Caps and Lids	0.1%	0.1%	2.1	1.1	0.1%	0.1%	128	1.9	1.1	147
Other Metal	5.1%	1.0%	66.2	22.6	1.4%	0.4%	543	140	3.4	1,124
Smoking	1.4%	0.3%	177.6	4.8	5.0%	0.6%	1,593	265	19.0	13,444
Cigarette Butts	0.0%	0.1%	4.2	1.7	1.4%	0.0%	1,159	109	11.8	12,417
Other Cigarettes, Tobacco, and Cigarette Products and Packaging	0.0%	0.0%	0.4	1.2	1.4%	0.0%	146	19	1.4	199
Cigarettes and Tobacco	0.0%	0.0%	7.8	5.0	0.1%	0.1%	44	17	0.5	118
Food Waste	0.2%	0.0%	102.8	38.2	1.1%	0.1%	426	122	27.1	3,915
Human or Pet Waste	0.1%	0.1%	1.7	1.2	0.1%	0.0%	10	14	0.0	0
Other Organics	0.1%	4.0%	184.7	50.1	0.0%	0.0%	739	149	16.8	2,124
Construction and Demolition Debris	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	0.0	0
CSL Waste	0.0%	1.6%	118.7	18.8	5.1%	0.6%	1,961	240	368.7	2,619
Hazardous Materials	0.1%	1.4%	114.7	19.4	1.1%	0.4%	1,161	149	16.8	2,124
Chemicals, Fuels and Other Incombustible Materials	0.1%	0.4%	8.0	3.0	0.8%	0.1%	224	135	2.6	227
Medical Waste	0.0%	0.0%	6.7	4.0	0.1%	0.0%	22	0.0	0.0	0
Medical Waste	0.1%	0.1%	0.8	0.7	0.2%	0.1%	131	198	0.8	140
Medical Waste	0.0%	0.0%	0.1	0.1	0.2%	0.0%	37	0.7	0.7	75
Explosives (Shave gel, fireworks or non-aerosolized latex)	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	1.2	1
Other Materials	55.1%	5.1%	683.1	61.2	31.8%	4.5%	12,216	1,307	384.8	24,301
Fishing and Maritime Items	0.2%	0.1%	2.0	1.3	0.1%	0.1%	53	13	0.9	9
Tires	0.0%	0.0%	44.1	30.7	0.0%	0.0%	6	6	0.1	11.5
Auto Rubber Products (Stumps, tire shreds or treads)	22.7%	4.4%	273.9	10.2	18.3%	1.3%	7,045	1,147	10.4	11,798
Vehicle Debris	0.0%	0.0%	288.1	31.9	0.1%	0.1%	1,197	147	18.1	1,816
Batteries, Electronics, and Small Appliances	0.1%	0.0%	1.1	0.5	0.1%	0.0%	21	18	1.5	15
Large Appliances	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	0.0	0
Household/Commercial Items/Office Items	1.0%	0.0%	12.4	5.5	0.0%	0.0%	189	19	25.5	424
Toilets, Clothing and Shoes	0.0%	0.0%	43.1	30.2	0.0%	0.0%	145	146	24.8	1,144
Toiletries and Personal Products	0.2%	0.1%	2.8	0.8	1.1%	0.0%	441	102	0.5	172
Batteries	0.0%	0.0%	0.0	0.0	0.0%	0.0%	6	0.1	0.1	2
Toys, Sports, and Recreational Items	0.1%	0.0%	0.7	0.4	0.1%	0.1%	40	10	2.7	55
Furniture, Carpets, and Bedding Items	0.0%	0.0%	0.0	0.0	0.0%	0.0%	0	0	2.8	10
Whole Bags of Mixed Trash	0.1%	0.0%	3.8	3.0	0.0%	0.1%	6	5	18.1	12
Miscellaneous Materials	0.1%	1.1%	11.0	10.8	0.1%	0.1%	116	71	1.7	1,211
Estimated Total (per mile)	100%		1,127.0	129.6	100%		18,518	1,578.8	173.68	21,518

Figure 105. Sample sorting, full sample (left), sub sample (right).



Goals

Collect granular data on functional categories

Get detailed counts instead of weights

Easy data collection process

Our Solutions

Use an applet for item-by-item data entry

Weigh total trash, alongside selected trash categories

Conclusion

The categories are highly adapted to ECCA's concerns and needs. We sacrificed some amount of compatibility with the WSDOT study to accomplish these goals.

ECCA Roadside Litter Study

Collection Protocol

Procedures, Sampled Areas & Field Stages

Highway 14 Corridor · Clark & Skamania Counties, WA

Why This Protocol Exists

Ensure data consistency across all sampled locations and volunteer teams, regardless of who collects on a given day.

Minimize risk and complexity during roadside collection by keeping categorization off-site.

Maintain methodological compatibility with the 2022 Washington Statewide Litter Study to enable meaningful regional comparisons.

8 Segments · 5 Highway Strata · 3 Ramps

5
Highway
Strata

3
Ramp
Segments

~3
Month
Accumulation

Accumulation Rule

Each segment must remain **uncleaned for ~3 months** before each collection event. This interval ensures statistically meaningful accumulation while not significantly delaying ECCA's ongoing cleanup commitments.

Segment IDs

H1-8 Camas North

H2-1 Camas South

H3-3 Washougal Xtra (S)

H4-10 Washougal North

H5-5 Washougal South

014Q101046 Vancouver S on-ramp

014S101194 Camas N on-ramp

014P101430 Washougal S off-ramp

The Protocol in Two Stages

1

Preparation

Before any roadside work begins

- Confirm ~3-month accumulation period has elapsed
- Assemble and check all materials
- Brief volunteers on boundaries & safety

2

Roadside Collection

Speed and safety; no sorting on-site

- Collect all litter within segment boundaries only
- Label every bag with Segment ID before filling
- Record large/bulky items on separate sheet

Before Leaving for the Field

Materials Checklist

- Collection bags — labeled by Segment ID
- Large-item recording sheet
- Portable scale
- Applet device (charged & tested) for data entry
- Segment boundary info (map links from sampling results table)
- Hi-vis vests, gloves, litter picker tools

Volunteer Briefing

Exact segment boundaries

Confirm start/end points for today's collection using Google Maps links.

Bag separation

Bags from different segments must be kept physically separate at all times — from collection through transport.

Safety requirements

Hi-vis vests must be worn. Follow all roadside safety procedures before stepping outside the vehicle.

Collect Fast, Stay Safe, Don't Sort

Boundaries

Collect only within designated start/end points confirmed by Google Maps links. Straying into adjacent areas changes the effective sample length and invalidates cross-segment comparisons.

Bag Labeling

Every collection bag must be labeled with the Segment ID. To prevent lost or damaged labels, bags should be labeled immediately after filling rather than before. Bags from different segments must be kept physically separate at all times

Large / Bulky Items

Items too large for standard bags (tires, lumber, construction debris) are documented on the large-item recording sheet: description, estimated size/weight, and likely source category.

Key Rule: No item-by-item categorization takes place at the roadside — bag everything, label it, and categorize at the sorting location.

Key Reminders for Every Collection Event



Wait the full ~3 months

Never collect a segment early — it breaks temporal comparisons.



Stay within boundaries

Google Maps links confirm exact start/end points.



Weigh before opening

Total segment weight is logged before any bag is opened.



Label bags after filling

Segment ID on every bag before the first item goes in.



No roadside sorting

All categorization happens at the central sorting location only.



Log three fields

Segment ID · Collection date · Total weight (consistent units).

ECCA Roadside Litter Study

Categorization Protocol

Why This Protocol Exists

Collect litter data the same way at every location.

Use the app to record each litter item one at a time.

Identify common litter types and where they may come from.

Create organized data that can be used for future studies.

The Categorization Process in Three Stages

1

Initial Weighing

Before
categorization
begins

2

Item-by-Item Categorization

Each item is
entered into the
app

3

Category Weighing

After categorization
is complete

STAGE 1: Before Categorization Begins

Information to Record

The first stage begins after the litter has been collected and transported to the sorting location

Volunteers record basic information about the sample.

- Segment ID
- Collection Date
- Total Litter Weight
- Total Plastic Weight (if available)

Volunteer Briefing

Why Record Weight First?

Recording this information first provides a baseline measurement for the sample.

It also helps ensure that all collected litter is included in the study.

Stage 2: Item-by-Item Categorization

Examine the Item

Volunteers examine each litter item individually.

For each item, they first identify what the item is.

Record Information in the App

Then, they enter the item information into the app, including its material, function, and other details.

Assign Categories

Each item is classified using:

- Material Category
- Functional Category
- Likely Source
- Brand Information (if available)
- Size Information

This process provides detailed information about the litter found in each sample area.

STAGE 3: Weigh Selected Categories

1

After Categorization

After all items have been classified, selected categories can be weighed separately.

2

Record Key Measurements

Current measurements include:

- Total Litter Weight
- Total Plastic Weight

3

Document Large Debris

Large or unusual items that do not fit easily into collection bags are recorded separately with additional notes.

Current focus: The main measurements of interest are total litter weight and total plastic weight.

Conclusions & Next Steps

What This Study Delivers

A replicable sampling design

8 segments (5 highway + 3 ramps) chosen via stratified random sampling, with defined weights for total litter estimation.

A consistent collection protocol

Three-stage field procedure — preparation, roadside collection, and initial weighing — standardized across all volunteer teams.

An item-level categorization system

Applet-based data entry covering material, function, likely source, brand, and size/fragmentation for every item.

Statewide comparability

Methodology aligned with the 2022 WA Statewide Litter Study, enabling future regional benchmarking.

Recommended Next Steps

1

Conduct the first collection round

Each sampled segment must be left uncleaned for ~3 months before collection begins.

2

Sort and categorize collected items

Use the applet to record all item characteristics at the central sorting location.

3

Investigate strata consolidation

With only one sample per highway stratum, combining strata where data supports it will improve analytical power.

4

Plan for multi-round comparison

Repeated collections across time are the path to understanding litter accumulation rates.